

Kern -- Project Architecture Brief

Confidential | Version 0.1.0.0 | June 2026

1. Overview

Kern is a privacy-first, fully offline Android toolkit for editing Office documents, spreadsheets, presentations, PDFs and CSV files. It operates with zero network permissions and writes files back to their original location without copying them into an app-owned folder.

The core design principle is that the user's files remain entirely under their control. Kern never uploads, syncs, or transmits any file content. It is released under the GNU Affero General Public License v3.0 (AGPL-3.0).

2. Supported Formats

Format	Library	Capabilities
Excel (.xlsx)	Apache POI	View, Edit, Export
CSV (.csv)	OpenCSV	View, Edit, Export
Word (.docx)	Apache POI	View, Edit, Export
PDF (.pdf)	MuPDF via Qyra JNI	View, Annotate, Edit, Export
PowerPoint (.pptx)	Apache POI	View, Edit, Export
EPUB (.epub)	TBD	View (Alpha)

3. Architecture

Kern uses a single-Activity Compose architecture with a NavHost routing between a file browser and individual editors. Files are accessed via Android's Storage Access Framework (SAF) and `MANAGE_EXTERNAL_STORAGE` permission, scoped to the Documents and Downloads directories.

The PDF engine is sourced from the Qyra open-source project (github.com/zParik/Qyra), which wraps the MuPDF C library in a Rust layer. Kern calls this layer via JNI/NDK. All other format handling is pure Kotlin/JVM using Apache POI and OpenCSV.

4. Release Plan Summary

Version	Stage	Description
0.1.0.0	Alpha	Project skeleton, CI, intent filters, file browser shell
0.1.1.0	Alpha	CSV view + edit + export
0.1.2.0	Alpha	Excel view + edit + export
0.1.3.0	Alpha	Word view + edit + export
0.1.4.0	Alpha	PowerPoint view + edit + export
0.1.5.0	Alpha	PDF via Qyra MuPDF JNI bridge
0.1.6.0	Alpha	Integration polish, dark mode, APK audit
1.1.0.0	Beta	Cross-format stability, undo/redo, large file handling
1.1.1.0	Beta	PDF feature parity: OCR, password protect, compress
1.1.4.0	Beta	Beta freeze, Play Store + F-Droid prep
R.1.0.0	Release	Public launch: Play Store + F-Droid

5. License

Kern is licensed under AGPL-3.0. All source code is publicly available. The PDF engine (MuPDF via Qyra) is GPL-3.0, which is compatible with AGPL-3.0 as a combined work. Both license headers are present in all bridge files.